

Program : Diploma in Architecture	
Course Code : 6181B	Course Title: Tendering and Valuation
Semester : 6	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4, T:0, P:0)	Periods per semester: 60

Course Objectives:

- To understand terminologies in contract and tender document and their significance.
- To know different types of contracts and their uses.
- To learn preparation of typical Tender documents for civil engineering work.
- To get acquainted with rent fixation and valuation of civil structures

Course Prerequisites:

Topic	Course code	Course name	Semester
Building construction methods		Building construction and materials	3

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Explain terminologies in contract and tender document and their significance	10	Understanding
CO2	Compare various types of contract and suggest the relevant type of contract for the given civil engineering work.	14	Understanding
CO3	Develop typical Tender document for the given civil engineering work.	16	Understanding
CO4	Make use of the standards to fix the rent and valuation of given civil structure.	18	Applying
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2					2	
CO3	2					2	
CO4	2					2	2

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Out come	Description	Duration (Hours)	Cognitive Level
CO1	Explain terminologies in contract and tender document and their significance		
M1.01	Discuss the terminologies in contract and tender document	3	Understanding
M1.02	List out methods for carrying out works	2	Understanding
M1.03	Explain methods for carrying out works	5	Understanding
Contents: Administrative approval, Technical sanction, budget provision, expenditure sanction. Methods for carrying out works- contract method, departmental method -rate list method, piece work method, day's work method, employing labors on daily wages basis.			
CO2	Compare various types of contract and suggest the relevant type of contract for the given civil engineering work.		
M2.01	Define contract and Indian Contract Act.	1	Understanding
M2.02	List the merits and demerits of different types of engineering contract	1	Understanding
M2.03	Explain FIDIC Conditions of contract.	4	Understanding
M2.04	Classify contractor on basis of financial limits	4	Understanding
M2.05	Explain the requirement of documents for registration of contractor	4	Understanding
	Series Test – I	1	

Contents:

Definition of contract, objects of contract, requirements of contract, overview of Indian Contract Act. - Types of engineering contract with advantages, disadvantages and their suitability - Lump sum contract, item rate contract, percentage rate contract, cost plus percentage, cost plus fixed fee, cost plus variable percentage and cost-plus variable fee contract, labor contract, demolition contract, target contract, negotiated contract, Engineering Procurement Construction Contract (EPC), Annuity Contract. - Introduction of FIDIC Conditions of contract. - Classification of contractor on basis of financial limits, Requirement of documents for registration of contractor. - Build Operate Transfer (BOT) Project, Hybrid Annuity contract, Operate Maintain and Transfer (OMT) contract, Operation & Maintenance contract (Introduction only).

CO3	Develop typical Tender document for the given civil engineering work.		
M3.01	Define tender and understand different types of tender	3	Remembering
M3.02	Discuss draft tender notice	4	Understanding
M3.03	List out the contents of tender documents	4	Understanding
M3.04	Explain tendering procedure and terms related to tender it.	5	Understanding

Contents:

Definition of tender, necessity of tender, types of tender - Local, Global, Limited. - E - Tendering System - Online procedure of submission and opening of bids (Technical and Financial). - Notice to invite tender (NIT)- Points to be included while drafting tender notice, Drafting of tender notice. - Procedure of submitting filled tender Documents (Two envelope system), procedure of opening tender, comparative statement, scrutiny of tenders, award of contract, letter of award. - Meaning of terms - Earnest Money Deposit (EMD), Performance Security Deposit, Validity period, corrigendum to tender notice and its necessity.

Terms related to tender documents - contract conditions- time limit, time extension, penalty, defective material and workmanship, termination of contract, suspension of work, subletting of contract, extra items, price variation clause(escalation), defect liability Period, liquidated Damages.

CO4	Make use of the standards to fix the rent and valuation of given civil structure		
M4.01	Define the purpose of valuation and role of valuer	3	Understanding
M4.02	Explain different types of value	3	Understanding
M4.03	Calculate depreciation	4	Applying
M4.04	Fix rent/lease of a property	4	Applying

M4.05	List precautions to be taken while making mortgage.	4	Understanding
	Series Test – II	1	
Contents: Introduction to Valuation - Definition and purpose of Valuation, role of valuer. Definition - Cost, Price and Value, Characteristics of Value, Factors Affecting Value. - Types of Value - Book Value, Scrap Value, Salvage Value, Speculative Value, Distress Value, Market Value, monopoly Value, Sentimental Value. Factors affecting value. - Depreciation, Obsolescence, Sinking Fund, Methods of Calculation of Depreciation - Straight Line Method, Sinking Fund Method, Constant Percentage Method. - Fixation of rent, Lease - types of lease, lease hold property and free hold property.			

Text / Reference:

T/R	Book Title/Author
T1	Datta, B. N., Estimating and Costing in Civil engineering, UBS Publishers Pvt. Ltd., New Delhi
T2	Raina, V. K., Construction Management and Contract Practices, Shroff Publishers & Distributers Pvt. Ltd.
T3	Rangawala, S. C., Estimating and Costing, Charotar Publishing House PVT. LTD., Gujarat
T4	Birdie, G. S., Estimating and Costing, Dhanpat Rai Publishing Company(P) Ltd., New Delhi
T5	Patil, B. S., Civil Engineering Contracts and Estimates, Orient Longman, Mumbai
T6	Chakraborti, M., Estimating and Costing, Specification and Valuation in Civil Engineering, Monojit Chakraborti, Kolkata.